

ADHAM SELIM

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EDUCATION

University of Twente

WO BSc Business Information Technology | Weighted Average: 7.85/10.0

Enschede, Netherlands

Graduated: July 2026

RESEARCH

MAYA: A Collaborative Multi-Agent System for Accelerating Academic Literature Review | [Link](#)

- Designed and developed an architecture for a multi-agent system to accelerate systematic literature search.
- Achieves retrieval speeds up to **twice as fast** and data extraction speeds up to **100 times faster** than a human.
- Data retrieval capability comparable to that of a human, as evidenced by a **63% Jaccard Similarity score**.
- Intended for all users of any field, by giving users freedom of choice between LLM and database(s).
- Incorporates an MCP server for standardized database communication with the tool.

EXPERIENCE

Student Intern

Ortes B.V.

Feb. 2026 – Apr. 2026

Deventer, Netherlands

- Designed and built an MVP for an interactive AI dashboard to be integrated with Ortes' system.
- Used PandasAI library to make the dashboard plug-and-play with most available LLMs.
- Wrote a feasibility report tailored for Ortes on how they can effectively incorporate the product into their workflow.

Co-Founder

Mochi

Feb. 2025 – Present

Enschede, Netherlands

- Co-founded Mochi, a virtual companion designed to ease stress and manage time, inspired by HCI principles.
- Secured entry for Mochi into the UT Challenge 2025, an entrepreneurship competition and advanced to the top 40 teams.
- Prototyped Mochi using tools such as Figma and basic HTML/React proof of concept.

Teaching Assistant/Mentor

University of Twente

Sep. 2024 – Jul. 2026

Enschede, Netherlands

- Mentored **100+ students** throughout multiple modules of the BIT/TCS courses.
- Coordinated & held weekly meetings for students to assess their progress and offer guidance.
- Graded hundreds of student assignments and multiple projects with a focus on constructive feedback

PROJECTS

Gramm3 - CALL: Teaching English to Arabic native speakers.

- Fine-tuned a Google Gemma Model using Unsloth with **QLoRA 4-bit quantization** to create a lightweight model.
- Designed the model for Computer Assisted Language Learning; specifically teaching English to Arabic native speakers.
- Achieved accuracy of **~80% on corrected sentences in ~90% of all test cases**.
- Scored **between 84-95%** when evaluated using BLEU and ROUGE 1,2, L and LSUM metrics.

Sign to Speech: Real-time ASL translation into speech and text

- Trained a **custom CNN model** on public datasets of ASL hand skeletons using **tensorflow keras**.
- Used mediapipe python library and camera/video input normalization for real-time ASL translation.
- Set up a docker container to run the model on a Raspberry Pi 5 for a portable device with a camera and touchscreen.

AnalystGPT: explain the effect of stock market metrics on analyst recommendations

- Trained a custom model on real-time financial data from the S&P 500, using the decision tree classifier.
- Predicts stock buy/hold/sell recommendations with **accuracy of 69.7%**.
- Relies on XAI concepts and the tool to analyze and explain the effect of various stock market metrics on analyst recommendations.

Note-Bridge Forum : Online interactive forum for music enthusiasts

- Built a website of an online forum using HTML, CSS & the Bootstrap framework for startup company "Note-Bridge" as a 8-week university project, with a team of 5 students
- Constructed a normalized (Postgre)SQL database schema of 13 tables with SQL-injection security.
- Created a full account creation and management system that allowed for collaboration and communication between users with different accounts.

TECHNICAL SKILLS

Tech Languages: Mostly used: Python, **Have worked with:** Java, SQL, JavaScript/TypeScript, HTML/CSS

Frameworks & Libraries: Pandas, FastAPI, PyTorch, Scikit-Learn, Tensorflow

Tools & Platforms: Git/GitHub, Docker, Cursor, Claude

Languages: English (Native), Arabic (Native)